FROM SEARCH ENGINES TO AI OVERVIEWS: A COMPARATIVE STUDY OF WEB CONTENT DISCOVERY

(A THREE ESSAY DISSERTATION)

by

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DEDICATION

To my precious daughter, Selma, whose laughter and love brightened even the most challenging days.

To my father, Ameer, and my mother, Amira, whose sacrifices, guidance, and unconditional love laid the foundation for all my achievements.

And to my beloved husband, Haytham, whose unwavering support and encouragement have been my greatest strength throughout this journey.

This work is as much yours as it is mine.

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ABSTRACT

Advancements in Artificial Intelligence are transforming how information is delivered in search environments by incorporating synthesized AI overviews alongside traditional SERP listings. Unlike conventional retrieval algorithms that return indexed website links, AI Overviews generate integrated summaries derived from multiple web sources, reshaping the user's informational experience and raising new challenges for platforms, SEO strategies, and digital information ecosystems.

Drawing from Media Richness Theory and Media Synchronicity Theory, this three-essay research investigates the extent, drivers, and consequences of divergence between AI Overviews and traditional search results. The first essay examines whether AI-generated results differ from SERP listings in terms of content selection and source overlap. Findings from 131 travel-related queries show an average URL overlap of only ~11%, indicating that AI-generated summaries draw from a comparably distinct source base.

The second essay applies Media Synchronicity Theory to explore why this divergence arises. Linguistic and structural analyses—including lexical diversity, cluster entropy, noun

diversity, Gini index, and text-to-HTML ratio—reveal a consistent pattern: AI Overviews favor convergence-oriented communication marked by cohesion and semantic focus, whereas SERP results support conveyance-oriented exploration through broader, more varied sourcing.

The third essay examines how different media formats shape cognitive processing outcomes. Using a between-subjects experimental design, participants were exposed to either AI-generated overviews, traditional SERP results, or a hybrid media. Results show that AI Overviews increase processing fluency, which reduces psychological distance and produces higher satisfaction and decision readiness. These findings suggest that AI functions not merely as an informational overlay, but as a cognitive restructuring mechanism that alters how users mentally engage with search-based information.

Collectively, this work advances theoretical understanding of AI-mediated information environments and offers practical implications for search platform design, content strategy, and the future of digital knowledge access.

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ESSAY 1: THE EVOLUTION OF SEARCH: EXPLORING HOW AI OVERVIEWS INFLUENCE SEARCH ENGINE OPTIMIZATION

INTRODUCTION

Search engines are a fundamental mechanism through which users retrieve information from the Internet. On average, users conduct 8.5 billion searches per day, or 99,000 per second. Additionally, for organizations seeking to leverage Google's search results, it is imperative to appear on the first page as it is estimated this captures 92% of traffic flow (Kathleen, 2024). These statistics underscore the significant financial stakes of changes in search result presentation. Search has become the digital gateway to knowledge, commerce, and communication. In an age where nearly every online journey begins with a query, search engines wield immense influence over what we see, believe, and buy.

The evolution of search engines reflects a broader narrative of digital transformation. From primitive keyword matching in early systems to the introduction of PageRank by Google, which prioritized link authority, the search landscape has continuously adapted to scale and complexity (Levene, 2011). More recently, the integration of machine learning and natural language processing has revolutionized how search understands user intent (Amer & Elboghdadly, 2024).

Since the inception of ChatGPT in 2022, many organizations have been leveraging GAIs to present enriched content to users. Search engines like Google and Bing have integrated large language models (LLM) to deliver summarized overviews of the search results of users' queries, along with URL references. GAI overviews, unlike traditional retrieval algorithms (Rogers, 2002) generate summaries of retrieved web sources (Lewis, Perez, Piktus, Petroni, Karpukhin, Goyal, Küttler, et al., 2020), reshaping the search experience by transforming search results' content into a quick, user-friendly overview.

GAI overviews bring new challenges, particularly in how source website content affects their inclusion and the broader implications for search platforms, web designers, and SEO professionals (Amer & Elboghdadly, 2024). For instance, organizations are designing webpages to meet user preferences and SEO optimization preferences for interactivity, responsive design, dynamic content, and seamless experiences requiring advanced technologies and intricate website structures. While this content may be appropriate for users, whether it is suitable for consumption by recent GAI is an important new research question: *How AI overviews Influence Search Results?* To answer this question we draw on Media Richness Theory (Daft & Lengel, 1984), which emphasizes users' preference for richer media forms, and argue that AI overviews could disrupt this dynamic, as GAIs tend to source content from websites with lower levels of media richness.

This essay investigates how the emergence of AI Overviews is reshaping traditional search engine optimization (SEO) strategies. It argues that AI overviews underlying mechanisms will affect the retrieval of websites in Search engines, with far-reaching implications for content creators, marketers, and users. We do so by collecting a dataset of URLs from Google's AI overviews and Search engine results for travel-related queries and comparing how much of the websites retrieved by AI overview are also retrieved by the traditional Search Engine (SE), which revealed minimal URL overlaps. These findings call for a redefinition of media synchronicity, urging organizations to adapt SEO strategies for both human and AI preferences.

This research contributes to a deeper understanding of how AI is not merely augmenting but transforming search. It connects information systems, communication theory, and artificial intelligence in a timely analysis of algorithmic mediation. The findings will inform content strategy, SEO practices, and platform governance, with broader implications for user autonomy, transparency, and information diversity. The objectives of this research are as follows:

- Quantify the overlap between AI-generated Overviews and traditional SERP URLs.
- Identify patterns in content types and domains retrieved by each system.

- Examine whether differences reflect fundamental shifts in how information is structured and retrieved.

The structure of this thesis is organized to systematically explore the evolving role of AI in search and its implications for SEO and communication. Introduction section introduces the research and provides background on the evolution of search engines, tracing their development from early retrieval models to the emergence of generative AI. Section 2 offers a comprehensive literature review on AI-powered search technologies and shifting SEO paradigms. Section 3 outlines the methodology and data collection approach used to quantify the overlap between AI Overviews and traditional search engine results. Section 4 presents the findings from Essay 1, focusing on the degree of content overlap and domain-level divergence. Finally, the last section provides insights and conclusions of the research.

BACKGROUND

The search engine landscape is undergoing significant change. In the past, retrieval-based algorithms like PageRank, Hummingbird, and others were used by search engines like Google to rank and media pertinent webpages as lists of clickable URLs (Ssemwogerere, Sajo, Mutwalibi, & Mzee, 2023). Given that a vast majority of user clicks have historically occurred on the first page of search results (Brian, 2025) and the impact of user behavior on search engine rankings (Ignite, 2022), Search Engine Optimization (SEO) is a significant field to be studied whose goal is to increase a website's visibility and the volume of organic traffic it receives. Direct retrieval is the foundation of traditional Search Engine Results Pages (SERPs). The search engine uses a combination of user behavior signals, page load speed, website authority (usually determined by backlinks), content relevance, and hundreds of other ranking factors to return a ranked list of relevant documents or websites when a user submits a query (T.-Y. Liu, 2009; Moz; Saab, 2023). Accordingly, the SERP serves as a carefully chosen directory of information sources that enables users to autonomously investigate various viewpoints, confirm the reliability of the sources, and choose the ones that best suit their information requirements (Roy, Maxwell, & Hauff, 2022).

However, this retrieval-based approach is drastically changing with the introduction of AI Overview systems, such as Microsoft's Bing AI Chat and Google's Search Generative Experience (SGE). AI Overviews synthesize and summarize information directly on the results page, eliminating the need to explore links to external sources for users to peruse. To create logical, narrative-like summaries that aim to answer the user's question comprehensively, these overviews make use of large language models (LLMs) and retrieval-augmented generation (RAG) techniques (Lewis, Perez, Piktus, Petroni, Karpukhin, Goyal, Küttler, et al., 2020).

While AI overviews may enhance speed and reduce cognitive load, they raise concerns about transparency, accuracy, and source diversity (Weise & Metz, 2023). Users may be unaware of how information is selected, what is omitted, and whether they are being exposed to biased or incomplete representations of a topic. Additionally, early evidence suggests that AI summaries is estimated to reduce click-through rates to source websites by over 30% (Austin, 2025), potentially risking the economics of open web publishing. This phenomenon has prompted the emergence of optimization strategies such as Generative Engine Optimization (GEO) (Aggarwal et al., 2024), which focuses on maximizing content visibility within AI-generated outputs, like those found in AI Overviews, content creators must now produce machine-readable, semantically rich, and high-authority content to be featured within generative outputs (Ziakis & Vlachopoulou, 2023) rather than solely focusing on traditional Search Engine Results Pages (SERPs). Meanwhile, users may experience cognitive overload, trust concerns, or reduced engagement due to diminished transparency and the limited diversity of perspectives (Aimilia & Despina, 2020; Lahlou, 2025).

Traditional SERPs and AI Overviews differ in several fundamental ways that have implications for how users access and evaluate information: (a) Synthesis vs. Retrieval: AI Overviews generate a synthesized response by aggregating content from multiple sources into a unified narrative, while traditional SERPs present discrete, independent documents that users must individually explore. (b) Research suggests that AI systems may exhibit biases, favoring content with high linguistic clarity, lower media richness (fewer multimedia elements), and greater semantic predictability (Weise & Metz, 2023). (c) User Interaction: Ttraditional SERPs demand active user engagement, requiring individuals to evaluate and navigate among multiple sources. In contrast, AI Overviews offer immediate synthesized responses that may reduce exploratory behavior and critical evaluation, potentially altering how users process and validate information.

From a communication theory perspective, AI Overviews shift the control of information sequencing, framing, and emphasis from the user to the algorithm—fundamentally altering the nature of the interaction. This shift aligns with Media Richness Theory, which posits that different media vary in their ability to convey rich, nuanced information (Daft & Lengel, 1984). While traditional SERPs allow users to navigate across multiple content formats and sources—engaging with rich media such as videos, interactive elements, and detailed documents—AI Overviews tend to distill content into concise, text-heavy summaries. As a result, the richness of the medium is reduced, constraining users' ability to explore diverse perspectives or engage deeply with multimodal content. In the context of Essay 1, this theoretical lens help guide us in our exploration of the types of sources cited by AI Overviews vs Traditional SE results, and thus we hypothesize:

H1: The set of URLs referenced in AI Overviews will significantly differ from the top 10 organic URLs returned by traditional search engine algorithms for the same queries.

For both theoretical and practical reasons, it is essential to determine whether there is a high or low source overlap between AI Overviews and search engine results. From a practical standpoint, it influences how website owners handle visibility in search environments that are increasingly controlled by AI. According to theory, it provides the essential starting point for investigating more complex mechanisms of algorithmic bias, platform design, and the changing function of web content as an input and output of intelligent systems, It also challenges the Media Richness theory, since users may now favor the fast summarized content instead of the media rich content. Although recent research has examined how AI agents produce content (AI, 2016; Baker et al., 2025; Bavarian et al., 2022; Kalla, Smith, Samaah, & Kuraku, 2023), little is known about the sources that are revealed when users ask questions in the real world. By contrasting the URLs chosen by AI Overviews with those returned by traditional search engine algorithms for the same queries, this research fills that gap. A crucial first step in comprehending the wider ramifications of AI-mediated information access is determining whether these systems favor the same or different sources.